
Learning Conway’s Game of Life with CNN-Transformer Hybrids: Architecture Scaling, Exposure Bias, and Activation Inductive Bias

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Abstract

1 We study how a CNN-Transformer hybrid learns the one-step transition rule of
2 Conway’s Game of Life (GoL) on a 40×40 toroidal grid. Starting from a model
3 trained with pure teacher forcing, we show that autoregressive rollout suffers
4 from exposure bias, that a straight-through-estimator (STE) multi-step scheme
5 fails due to a systematic gradient bias, and that scheduled sampling resolves this
6 cleanly. We further show that once a model saturates its capacity, scaling the
7 training set five-fold *regresses* tail-case accuracy in a way invisible to aggregate
8 validation metrics, while scaling the model five-fold instead eliminates essentially
9 all remaining errors. Motivated by recent work on activation-function inductive
10 bias for cellular automata, we independently reproduce the finding that a learnable
11 second-degree polynomial activation lets a 25–34 parameter CNN learn GoL
12 reliably where ReLU fails, and report a negative result transferring this activation to
13 our larger multi-step model. Finally, we show that sinusoidal positional encodings
14 yield empirically more translation- equivariant patch embeddings than learned
15 ones.

16 1 Problem and architecture

17 GoL is a deterministic cellular automaton: each cell’s next state depends only on its own state and its
18 8-neighbor live count (B3/S23). We treat one GoL step as image-to-image prediction on a periodic
19 40×40 grid. Our core architecture is a CNN-Transformer hybrid: two circular-padded convolutions
20 extract local, translation-equivariant features respecting the torus topology; these are cut into non-
21 overlapping 4×4 patches, linearly projected, combined with a *learned* absolute positional embedding,
22 passed through a Transformer encoder for global mixing, and projected back to per-cell logits. GoL
23 is a natural stress test for learned representations: prior work shows plain feed-forward networks
24 struggle to learn it reliably at all [Springer and Kenyon, 2021], and the Transformer stage [Vaswani
25 et al., 2017] trades the CNN’s clean equivariance guarantee for long-range context (Section 3). This
26 V4 design is itself the fourth step of an earlier architecture search, each step fixing one identified
27 issue: V1 (avg-pooled tokens, learned positional embedding, zero-padding) $\rightarrow$ V2 (swaps in a *fixed*
28 *sinusoidal* positional embedding to remove a right-edge rollout artifact) $\rightarrow$ V3 (swaps avg-pooling
29 for a lossless flatten-and-project tokenization, reverting to a learned embedding) $\rightarrow$ V4 (swaps
30 zero-padding for circular padding, fixing incorrect boundary neighbor counts). V5 onward (Section 2)
31 all train this same V4 architecture under different regimes and sizes; V2 reappears in Section 3 as the
32 point of comparison for positional-embedding geometry.

Ver.	Key change	Val. loss	Prec./Rec.	Status
V1	Baseline: avg-pool tokens, learned PE, zero-pad	— ^b	—	Edge artifacts; rollout freezes to a blob
V2	Learned PE → fixed sinusoidal PE	— ^b	—	Unstable training; no rollout improvement
V3	Avg-pool → flatten+linear tokenization	0.021	—	Fixes avg-pool bottleneck; first realistic rollouts
V4	Zero-padding → circular padding	0.226	100% / 94.3%	Fixes boundary bug; base arch. for V5+
V5	Single-step teacher forcing	0.201	99.4% / 99.7%	Collapses under rollout
V6	+STE multi-step	0.234 ^a	92.7% / —	Diverges
V7	STE → scheduled sampling	0.201	99.7% / 100%	Stable
V8	+3× high-density upsample	0.201	100% / 99.9%	Best per-grid (291K params)
V9	+5× all data	0.201	99.7% / 99.6%	Regressed tail errors
V10	5.15× model size	0.199	100% / 100%	Zero errors , 2,500/2,500

Table 1: Version summary, spanning the V1–V4 architecture search (Section 1) and the V5–V10 training-regime search (Section 2). ^aV6’s best epoch before divergence. ^bV1/V2 reported only aggregate val accuracy (89.4% and 84% respectively), not loss or a precision/recall split.

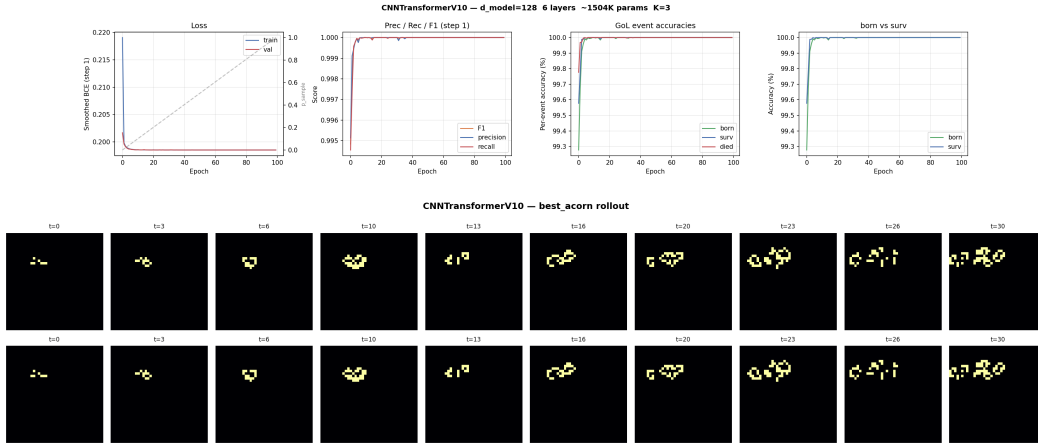


Figure 1: Top: V10 training curve. Bottom: a 30-step autoregressive rollout of the *acorn* methuselah — a 7-cell seed that grows into a large, long-lived chaotic pattern — true trajectory (top row) vs. V10’s autoregressive prediction (bottom row); V10 tracks the true trajectory exactly at every shown step, and makes zero errors across 2,500 held-out grids overall.

2 Training evolution: two failure modes and their fixes

Exposure bias. A model trained with pure single-step teacher forcing (V5) scores > 99% precision/recall on held-out single steps but collapses under autoregressive rollout, since it never sees its own (imperfect) outputs during training. A straight-through-estimator (STE) multi-step scheme intended to fix this instead **diverges** after epoch 5, which we trace to a systematic gradient bias from the STE’s hard threshold. Replacing STE with *scheduled sampling* [Bengio et al., 2015] — gradually increasing the probability of feeding the model its own previous prediction — fixes this cleanly and stably (V7).

A hidden data-scaling regression. Upsampling high-density configurations 3× (V8) improves per-grid accuracy further (291K params). Naively scaling the *training set* 5× (V9) leaves aggregate validation metrics statistically unchanged, but a density-stratified failure search (500 grids at each of five densities, fixed seed across models) reveals V9’s error count at density 0.80 is nearly **seven times worse** than V8’s (22,190 vs. 3,358 summed false positives/negatives) — invisible in aggregate metrics alone. Instead of more data, we scale the *model*: V10 (5.15× V8’s size, 1.5M params, trained from scratch) achieves **zero errors** across all 2,500 test grids at every density (Table 1, Figure 1). The lesson: once capacity saturates, more data can silently trade average-case for tail-case performance, detectable only by per-subgroup error auditing, not validation loss.

50 3 Embedding geometry: sinusoidal vs. learned positional encoding

51 A CNN feature map is a *physical* embedding: convolution’s translation-equivariance means shifting
52 the input shifts the feature map identically. A Transformer’s patch embeddings, after a learned
53 absolute positional embedding, have no such guarantee — each position slot is an independently
54 learned vector unrelated to its neighbors. We tested this empirically on two trained models differing
55 in positional encoding: one using a fixed 2D sinusoidal encoding (V2, whose angle-addition identities
56 give closed-form near-equivariant structure) and one using a learned absolute positional embedding
57 (V8). Measuring cosine similarity between patch embeddings of a grid and of translated copies of it,
58 V2’s post-Transformer embeddings stay far more similar under translation (mean cosine similarity
59 0.997) than V8’s (0.890), consistent with the theoretical prediction (Figure 2, top). We note this is
60 not a fully isolated ablation of positional encoding alone: V2 and V8 sit three architectural changes
61 apart in the V1–V4 progression (Section 2), and besides the positional embedding itself, V2 uses
62 avg-pooled tokenization and zero-padding versus V8’s flatten-and-project tokenization and circular
63 padding. Either confound could independently affect translation-sensitivity; disentangling them
64 would require a matched ablation that varies only the positional embedding on an otherwise-fixed V4
65 backbone, which we have not yet run.

66 4 Independent verification: polynomial activations for GoL

67 Motivated by a recent claim that a learnable second-degree polynomial activation $\phi(x) = w_0 + w_1x +$
68 w_2x^2 lets a minimal CNN learn GoL reliably where ReLU fails almost completely [Ahmed and
69 Davis, 2026], we independently reproduced this rather than taking it on faith. We built two CNNs
70 matching the paper’s minimal architecture (two 1×1 convolutions plus one 3×3 circular-padded
71 convolution), **identical except for the activation function**, matching the paper’s reported parameter
72 counts exactly (ReLU: 25 params, polynomial: 34 params). Trained single-step, teacher-forcing, over
73 10 random seeds: ReLU converges to perfect validation accuracy in only 1/10 seeds (mean final
74 accuracy 89.4%), while the polynomial activation converges in 10/10 seeds (100.0% mean accuracy),
75 confirming the claim on our own toroidal GoL setup (Figure 2, bottom). We then attempted to transfer
76 this activation into our larger multi-step model (same size as V8) by replacing GELU with polynomial
77 activations in the CNN stage; this **failed**, diverging sharply after epoch 5 under scheduled sampling,
78 likely because the unbounded w_2x^2 term compounds through the autoregressive feedback loop in
79 a way it cannot in a single-step setting. Candidate fixes (a bounded/saturating variant, a separate
80 learning rate for polynomial coefficients) were identified but not yet pursued.

81 5 Conclusion

82 Three practical lessons emerge. First, autoregressive exposure bias is best fixed with scheduled
83 sampling rather than a hard differentiable relaxation (STE), which can introduce its own destabilizing
84 gradient bias. Second, aggregate validation metrics can hide severe, density-specific regressions;
85 per-subgroup error auditing is necessary once a model is scaled with more data rather than more
86 capacity. Third, an activation function’s inductive-bias advantage demonstrated at minimal scale and
87 single-step training does not automatically transfer to a larger, multi-step, autoregressively-trained
88 model — the interaction between an activation’s stability properties and the training regime matters
89 as much as the activation itself.

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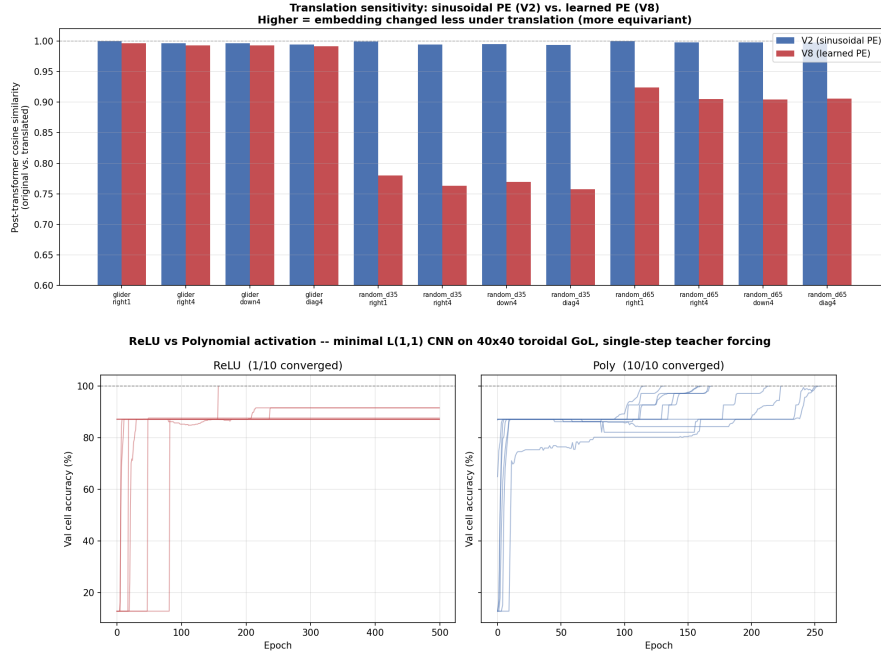


Figure 2: Top: post-Transformer embedding cosine similarity under translation, sinusoidal PE (V2) vs. learned PE (V8); higher means less change under translation, i.e. more equivariant. Bottom: validation accuracy over training, 10 seeds per activation, on a minimal 25/34- parameter CNN; polynomial activation converges reliably where ReLU does not.

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